

Wildfire burn severity and post-wildfire time impact mechanical and hydraulic properties of forest soils

Ryan A. Ramirez^{a,b,c}, Woojae Jang^a, Tae-Hyuk Kwon^{a,*}

^a Department of Civil and Environmental Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea

^b Department of Civil Engineering, University of Santo Tomas (UST), Manila 1008, Philippines

^c Division of Earth and Space Sciences, National Research Council of the Philippines (NRCP), Taguig 1631, Philippines

ARTICLE INFO

Keywords:

Forest wildfire
Burn severity
Post-wildfire time
Shear strength
Saturated hydraulic conductivity
Uljin county

ABSTRACT

Vegetation contributes to the overall slope stability and is recognized as an environmentally friendly nature-based solution. Wildfires burn and denude vegetated slopes, thus increasing the risk of shallow landslides and debris flows. However, little attention has been given to assessing the effects of burn severity and the time elapsed since a wildfire on the hydro-geomechanical properties of burned slopes. This study performed a series of standard laboratory tests to evaluate the shear strength and saturated hydraulic conductivity (k_{sat}) of forest soils collected from moderate-low (ML) burned, moderate-high (MH) burned, and unburned (UB) test plots. The plots were sampled one, four, and six months after the March 2022 wildfire in Uljin County, South Korea. The results show that the continuous deterioration of roots highly depended on the burn severity. The root biomass of ML- and MH-burned soils was consistently lower than those of the UB soils. The root deterioration reduced the shear strength of the soils temporally. The burned soil's cohesion intercept was 1.80–2.30 times lower than that of the UB soil six months post-wildfire, with the friction angle unaffected. One- and four-months post-wildfire, k_{sat} of the burned soils was 1.22–3.15 times lower than the UB soil. Such lowered k_{sat} was due to the fine ash-clogged pores and hydrophobic layers beneath the soil surface. However, six months post-wildfire, the burned soils' k_{sat} increased by approximately twice that of the pre-wildfire condition because of macropore flow passages formed by impoverished roots. The appreciation of sand fraction, depreciation of fines content, and weakening of hydrophobicity over time have also emphasized their role in the temporal shifts in the properties of the ML- and MH-burned soils. The documented results herein can be incorporated into rainfall infiltration and stability analyses of wildfire-affected slopes, landslide susceptibility mapping, and mitigation measures design.

1. Introduction

Wildfires burn and denude forested and (or) vegetated slopes relatively quickly over vast areas and cause considerable physico-chemical changes in shallow soils. Post-wildfire terrain conditions manifest different hydrogeological cycles due to reduced evapotranspiration associated with vegetation removal (Abbate et al., 2019). Additionally, loss of vegetation and litter due to wildfires reduces rainfall interception and rainwater infiltration, which thus causes increased surface runoff discharge and facilitates shallow soil erosion (Vieira et al., 2018). Shallow landslides, particularly runoff-generated debris flows, are more prone to occur following a wildfire during and (or) immediately after a high-intensity rainfall event. Some of the examples are the post-wildfire debris flow events that occurred in New Mexico (Cannon et al., 2001),

California (Cui et al., 2019; Kean et al., 2019) and Colorado (Palmer, 2022), the United States of America and in Australia (Nyman et al., 2011). Post-wildfire effect on burned shallow soils depends on several factors, including burn severity (or fire intensity), vegetation and soil types, fire frequency, and the time elapsed since the most recent wildfire (Moody et al., 2008).

Elevated temperatures due to prolonged wildfires can significantly alter soils' biological, chemical, physical, hydrological, and mechanical properties at shallow depths (Inbar et al., 2014; León et al., 2015; Vieira et al., 2018). For instance, exposure to high temperatures can cause aggregation of soil grains and result in a loss or gain in mass for fine-grained soils at shallow depths and the formation of a hydrophobic topsoil layer (Lei et al., 2022; Peduto et al., 2022a). Furthermore, high heat burns vegetation above ground, and the buried roots decay and, in

* Corresponding author.

E-mail address: t.kwon@kaist.ac.kr (T.-H. Kwon).

<https://doi.org/10.1016/j.geodrs.2024.e00856>

Received 2 June 2024; Received in revised form 29 August 2024; Accepted 31 August 2024

Available online 2 September 2024

2352-0094/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

the long term, reach death. Accordingly, such changes can alter water infiltration into soils and the shear strength of soils, associated with the presence of hydrophobic layers and degradation of plant root systems (Lei et al., 2022; Peduto et al., 2022b). Therefore, after wildfires and before vegetation regrows and soil recovers to its pre-wildfire status, the slopes would be highly vulnerable to shallow landslides and debris flows when exposed to heavy rainfall. However, there is still little attention and limited effort into quantitatively assessing the geomechanical and hydraulic properties of burned slopes immediately after a wildfire and how these properties change over time (Lei et al., 2022; Peduto et al., 2022a; Peduto et al., 2022b). Notably, the effects of burn severity and post-wildfire time on the engineering properties of shallow soils remain poorly understood.

This study explores temporal evolutions of the engineering properties of root-soil matrices that experienced wildfires with different burn severity. This study selected three sites with different burn severity exposed to the Uljin wildfire that occurred in 2022 in Uljin County, South Korea. A series of experiments was carried out to examine the temporal variations in physical, geotechnical, geomechanical, and hydrological properties of the natural soil samples from these sites one, four, and six months after the wildfire event, which includes the grain size distribution, root biomass content, water repellency, drained shear strength and saturated hydraulic conductivity. The presented results provide insights into to what extent the engineering properties are affected by wildfires and post-wildfire time and how slopes subjected to wildfires should be managed.

2. Materials and methods

2.1. Description of study area and wildfire occurrence

The study region is in Uljin County, North Gyeongsang Province,

Republic of Korea, about 330 km southeast of Seoul (Fig. 1). Uljin County has a humid continental, severe dry winter, and hot summer climate. Data collected from the Uljin Meteorological Observatory of the Korea Meteorological Administration (Korea Meteorological Administration KMA, 2022) for 1999–2020 indicate that the region's average annual temperature is 12.8 °C. Additionally, the average temperature in January, the coldest and driest month, and August, the warmest month, is 1.4 °C and 23.9 °C, respectively. The annual precipitation is roughly 1182 mm. Half of the precipitation falls in summer (from July to September). On the other hand, precipitation in February and April, when the frequency of forest wildfire is the highest, is roughly only 169 mm. Most landslides occur from July to September when precipitation is exceptionally high.

The study region sits on the Yeongnam Massif, located between the Ocheon Belt and Taebaeksan Basin in the north and the Gyeongsang Basin in the south (Oh et al., 2022). The geology in Uljin county is Precambrian granite gneiss. Plants of northern and southern provinces appear together due to topographical features connected to the high mountains. The forest landscape is dominated by Korean red pine (*Pinus densiflora* for. *Erecta*) forest and Mongolian oak (*Quercus mongolica*) forest (Jung et al., 2017).

On March 4, 2022, around 11:17 a.m., a wildfire in eastern coastal areas of Uljin County occurred, spreading to the vicinity of the mountain peak. Sentinel-2 Multi-Spectral Instrument (S2 MSI) optical image acquired on March 5, 2022, captured the wildfire in Uljin County and the neighboring city of Gangneung, as shown in Fig. 1b. Due to strong winds and dry weather, the wildfire spread north to Samcheok, Gangwon Province. Approximately 3300 ha of woodland was under the influence of the wildfire, according to the Korea Forest Service (Korea Forest Service KFS, 2022), the largest forest area to be affected by a single wildfire in the past decade. The wildfire burned at least 50 homes and led to the evacuations of about 4000 people in Uljin County and about

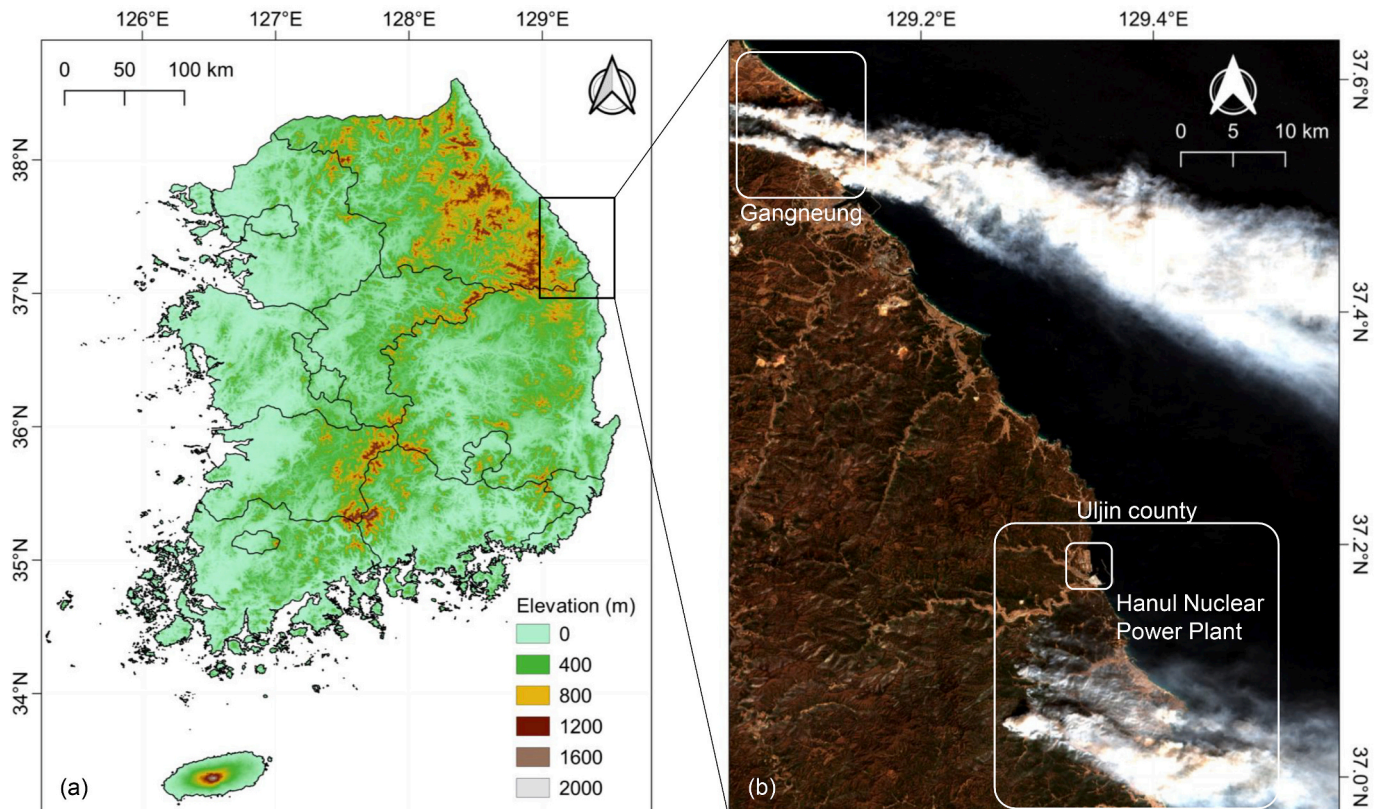


Fig. 1. (a) Map of the Republic of Korea showing the overview of the location of the study area and (b) Sentinel-2 Multi-Spectral Instrument (MSI) optical image of the wildfire in Uljin county and Gangneung, which started on March 4, 2022.

1000 people in Samcheok City, with no casualties reported. The primary fire was extinguished on March 15, 2022.

2.2. Soil burn severity estimation

This study used the optical image products acquired by cloud-free S2 MSI Level 2 A on February 23, 2022 (pre-fire image) and March 15, 2022 (post-fire image) to estimate the soil burn severity of the area of interest (AOI) under investigation. S2 MSI optical images are freely distributed through the Copernicus Programme under the European Space Agency (ESA) and the European Commission (EC) joint initiative. S2 MSI data are regularly acquired every five days (i.e., with two S2 satellites) and provide 13 multi-spectral bands with a spatial resolution of 10–60 m. The Sentinel Application Platform (SNAP), a suite of dedicated Sentinel product toolboxes freely distributed by ESA, was used to process the S2 MSI optical images. The soil burn severity of the AOI was estimated based on the relativized burn ratio (RBR) as follows (Parks et al., 2014; Quintano et al., 2018):

$$RBR = \frac{dNBR}{NBR_{pre-fire} + 1.001} = \frac{NBR_{pre-fire} - NBR_{post-fire}}{NBR_{pre-fire} + 1.001}, \quad (1)$$

$$NBR = \frac{NIR - SWIR}{NIR + SWIR}, \quad (2)$$

where $dNBR$ is the delta normalized burn ratio. The normalized burn ratio (NBR) is calculated based on the near-infrared (NIR) and the short-wave infrared ($SWIR$) spectral bands, as shown in Eq. 2, obtained from S2's multi-spectral imaging sensor. Table 1 lists the soil burn severity classes based on the range of $dNBR$ values (Key and Benson, 2006). Fig. 2a illustrates the soil burn severity map of the study area in Uljin county. The soil burn severity map provided the locations of the AOIs investigated in this study and further confirmed through field survey (i.e., test plots; Fig. 2b).

2.3. Field survey

The AOI was visited three times, one month, four months, and six months after the wildfire occurrence. One month after the wildfire, an aerial survey was first carried out using a DJI Inspire 2 drone to confirm the burning status of the selected test plots based on the soil burn severity map. For example, Fig. 2c and d show the aerial views of the AOI from one month to six months after the wildfire. After the drone survey, the ground surface condition, including ash deposits, was inspected on-site to confirm whether or not the selected test sites were exposed to the wildfire. Afterward, soil samples from three test sites with different soil burn severity, as marked in Fig. 2b and summarized in Table 2, were collected from the burned area one, four, and six months after the wildfire. These test sites with different levels of soil burn severity are referred to as the unburned site (UB), the site with moderate-to-low severity (ML), and the site with moderate-to-high severity (MH). The unburned site (UB) was used as a reference and control test site for comparison. The test sites for UB, ML, and MH burned soil samples were situated close to each other to ensure their

Table 1
Soil burn severity classification based on $dNBR$ (adopted from Key and Benson, 2006).

$dNBR$ range	Soil burn severity class
< -0.251	Enhanced high post-fire regrowth
-0.250 to -0.101	Enhanced low post-fire regrowth
-0.100 to 0.099	Unburned
0.100 to 0.269	Low severity
0.270 to 0.439	Moderate-low severity
0.440 to 0.659	Moderate-high severity
> 0.660	High severity

Note: $dNBR$ = delta Normalized Burn Ratio.

similar environmental conditions, such as slope and altitude, and minimize possible impacts of environmental factors on root distribution and properties. Also, these sampling locations had convenient access to a major thoroughfare.

2.4. Soil sampling

A 6×1.2 m test plot was chosen in each burn severity site for soil sampling, and each test plot was divided into 45 square sampling points with a regular interval (0.4×0.4 m; 15 square sampling points per visit), as described in Fig. 3. The sampling location in the following months (four and six months after the wildfire) was the same as in the previous month. Soil sampling was carried out with a cylindrical tube with a height of 150 mm and a diameter of 75 mm (i.e., a height-to-diameter ratio of 2), fabricated either with brass or aluminum alloy (AL6061). Relatively undisturbed samples were retrieved with sampling tubes after removing the charred organic layer from the surface (Fig. 3). When retrieving soil samples, the interior of the sampling tubes was coated in petroleum jelly to prevent preferential annular flow along the interface between the soil and the tube wall for saturated hydraulic conductivity measurement. Fifteen undisturbed samples per each burn severity test site were collected in each sampling period. The index and geotechnical parameters, including the bulk density, particle size distribution, specific gravity of solids, saturated hydraulic conductivity, and shear strength, were determined using those undisturbed samples. In addition, approximately 2 kg of disturbed soil samples were collected from a depth between 0 and 5 cm using an air-tight plastic bag to estimate the soil moisture content. All disturbed and undisturbed soil samples were stored in an air-tight condition at room temperature for various laboratory experiments. Table 3 summarizes the test program adapted in this study for the soil samples collected from the three test plots post-wildfire.

2.5. Water repellency test: Water droplet penetration time (WDPT) test

Additionally, before sampling and after removing the charred organic layer, a water droplet penetration time (WDPT) test was conducted near each plot's surrounding perimeter to evaluate the soil's water repellency in different burn severity areas (Glenn and Finley, 2010). This WDPT test was carried out on the ground surface five times (i.e., one in each corner and one in the center) for each 0.4×0.4 m subplot (Fig. 3). Using a pipette, a droplet of deionized water (i.e., volume = 1.7 mm^3) was dropped from 1 cm high above the soil surface, and the infiltration time until the water droplet disappeared was recorded. The mean value of the infiltration times of the five droplets was determined for each subplot, and this WDPT test was repeated for each visit to the UB, ML, and MH sites.

Moreover, we measured the WDPT and the contact angle (β) with laboratory specimens before testing to cross-examine the in situ WDPT results (Nuraje et al., 2013). A digital photo of a droplet contacting the soil surface was captured by using a digital camera (Canon EOS 200D II), and the contact angle β was estimated by an ImageJ plugin software Drop Snake. Accordingly, the WDPTs measured on the soil specimens in the laboratory were recorded and compared to in situ WDPTs.

2.6. Soil index property testing

All samples collected per field sampling effort from each site were tested in the laboratory. Table 3 summarizes the test program conducted in the study. The moisture content (w) of the disturbed samples from each bag was determined using the oven-drying method (ASTM D2216–19). The average w of nine replicates (i.e., three replicates from each bag) was determined as the representative w of each test plot during the three sampling periods. Using the relatively undisturbed samples, the apparent bulk density (or moist density; ρ) was estimated as the ratio of the moist weight of the specimen to its volume (ASTM

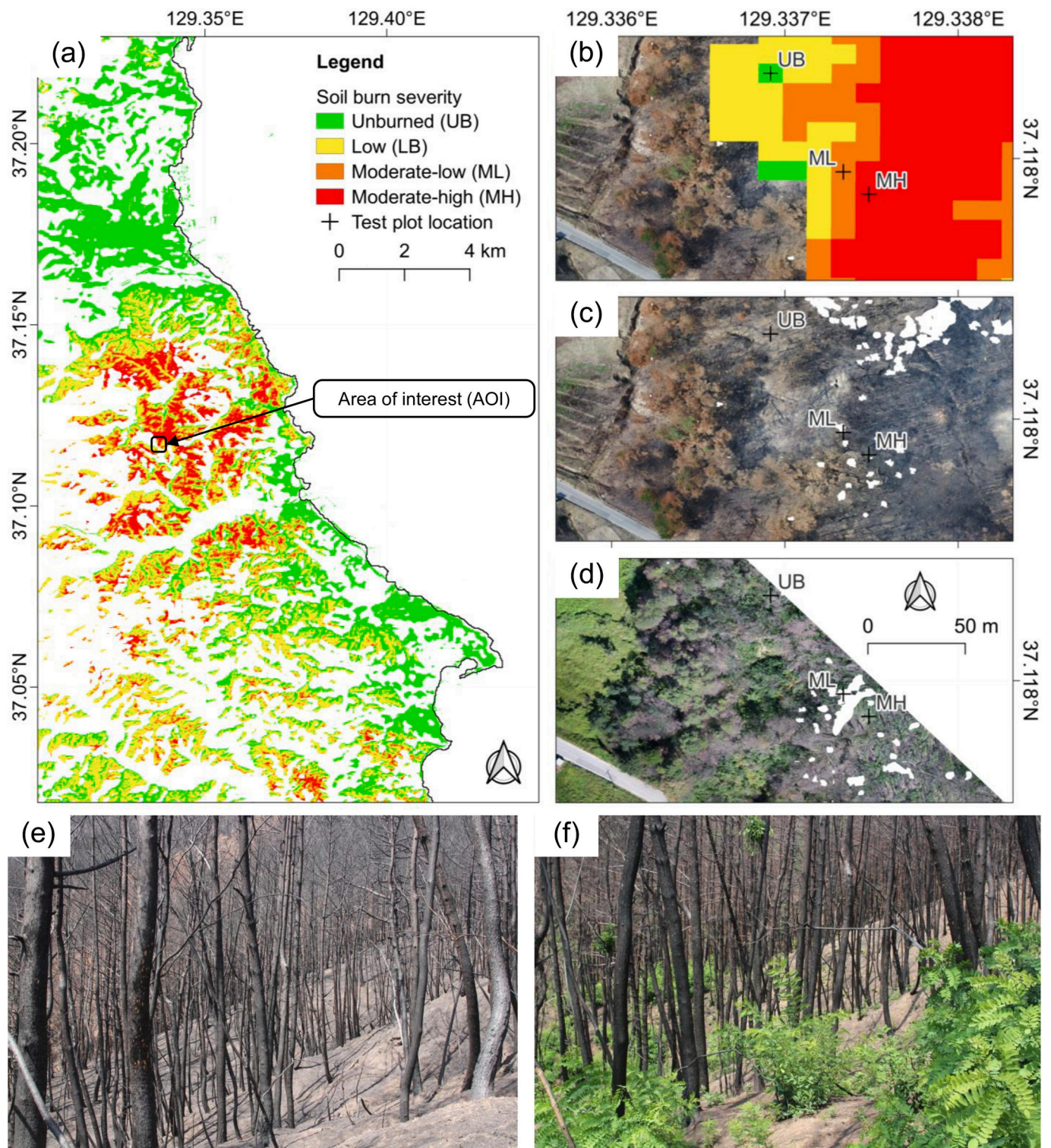


Fig. 2. (a) The soil burn severity map of the March 2022 wildfire in Uljin country, Republic of Korea. (b) The soil burn severity of the test plots in the area of interest (AOI). The aerial photos of the AOI taken during the first (i.e., one-month post-wildfire) and third (i.e., six months post-wildfire) sampling efforts, respectively, are shown in sub-figures (c) and (d). Ground conditions of the moderate-high-burned slopes (e) one month and (f) six months post-wildfire.

D7263–21). The average ρ of 15 specimens was chosen for the ρ of each test plot during the three sampling periods. Subsequently, the apparent dry density (ρ_D) was also calculated.

The particle size distribution and the specific gravity of soil solids (G_s) passing the No. 4 sieve were determined by using the sieve analysis (ASTM D6913–04) and the pycnometer analysis (ASTM D854–14), respectively, after conducting the constant head permeability tests with

the undisturbed saturated specimens. The liquid and plastic limits were determined following Casagrande's cup method (ASTM D4318-17e1). The particle size distribution results were used to classify the soil specimens according to the Unified Soil Classification System (USCS; ASTM D2487–06) and the World Reference Base for Soil Resources (WRB: [IUSS Working Group WRB, 2022](#)). Before estimating each soil index property (i.e., soil moisture and gradation), roots were carefully separated from

Table 2
Geographical coordinates of test sites with different burn severity.

Burn severity	Geographical coordinates		Slope angle, (°)	Slope aspect
	Longitude, (°)	Latitude, (°)		
Unburned (UB)	129.336849	37.117538	38.7	South- facing
Moderate-low (ML)	129.337070	37.117538	17.2	South- facing
Moderate-high (MH)	129.337188	37.117434	21.7	South- facing

the soil particles and used to measure the root biomass, as discussed in [Section 2.9](#).

2.7. X-ray computed tomography imaging and hydraulic conductivity test

X-ray computed tomography (X-ray CT) imaging was conducted on undisturbed tube samples (i.e., those subjected to the constant head hydraulic conductivity test). In this study, the X-ray CT imaging used a voltage of 150 kV and a current of 1000 μ A with a copper filter to acquire high-resolution images. As a result, the entire height of a soil tube was imaged with a stack of 350–400 sliced images, and each sliced image was composed of 1024×1024 pixels with a voxel size of $93.3 \times 93.3 \times 78.7 \mu\text{m}^3$. The obtained images were used to visualize the presence of roots and the internal structure of the specimens. [Fig. 4](#) shows representative cross-sections of one-month post-wildfire soil specimens from each burn severity site. Gray levels obtained in the CT slice correspond to different X-ray attenuation. Denser or less porous materials like grains are brighter, while air or water-filled pores or roots are

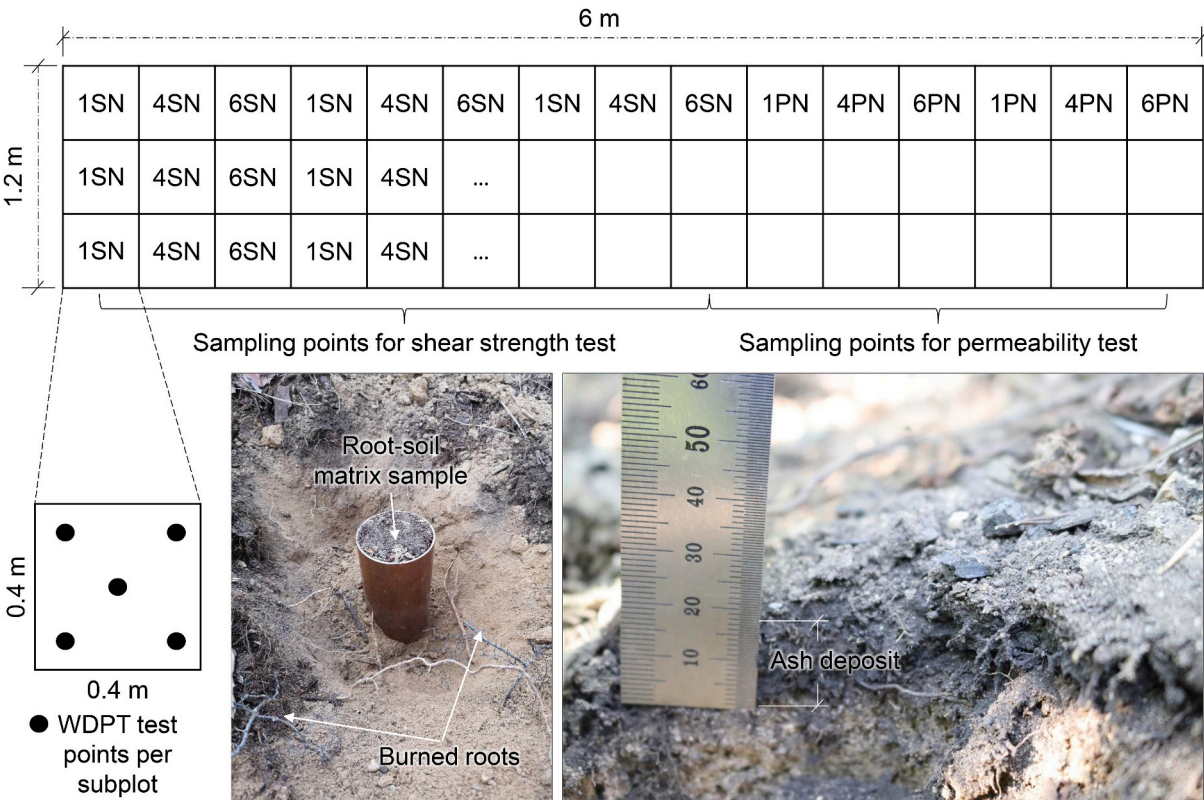


Fig. 3. Schematic diagram of the layout of the sampling test plots for the unburned and burned sites. Each test plot was divided into 45 square sampling points measuring 0.4×0.4 m. The numbers 1, 4, and 6 indicate the post-wildfire time of one month, four months, and six months. The letters *S* and *P* indicate the type of tests to which the samples will be subjected in the laboratory. *S* stands for shear strength testing, and *P* stands for permeability testing. The letter *N* represents the sample number. For the shear strength test, $N = 1\text{--}9$, and for the permeability test, $N = 1\text{--}6$.

Table 3
Test program for soil samples.

Condition	Number of samples or replicates used					
	Soil moisture (D)	Dry density (U)	Particle size distribution (D)	Specific gravity (D)	Shear strength (U)	Hydraulic conductivity (U)
1 month, UB	9	15	6	6	9	6
1 month, ML	9	15	6	6	9	6
1 month, MH	9	15	6	6	9	6
4 months, UB	9	15	6	6	9	6
4 months, ML	9	15	6	6	9	6
4 months, MH	9	15	6	6	9	6
6 months, UB	9	15	6	6	9	6
6 months, ML	9	15	6	6	9	6
6 months, MH	9	15	6	6	9	6

Note: UB = unburned soil; ML = moderate-low-burned soil; MH = moderate-high-burned soil; D = disturbed sample; U = undisturbed sample.

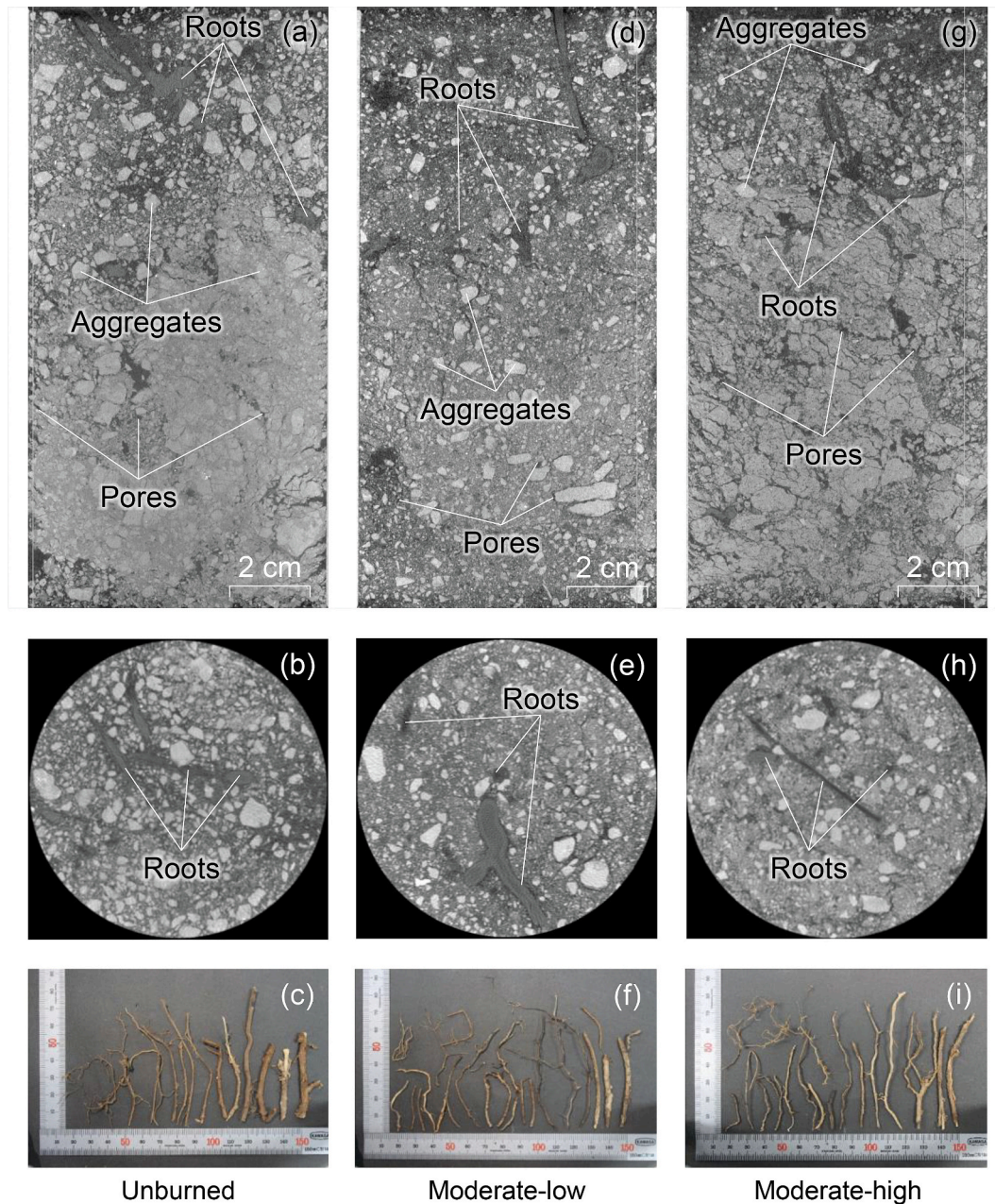


Fig. 4. X-ray computed tomography (XCT) image of representative samples collected from the (a, b) unburned site, (d, e) moderate-low-burned site, and (g, h) moderate-high-burned site one-month post-wildfire. The images show the two-dimensional vertical (a, d, g) and horizontal (b, e, h) cross-sections of the samples collected one-month post-wildfire. The photos of oven-dried roots retrieved from the (c) unburned soil, (f) moderate-low-burned soil, and (i) moderate-high-burned soil samples.

shown as darker spots. In X-ray CT images, coarse roots were visible, and these roots have mostly circular shapes, differentiating them from the soil particles or other darker areas. The images also reveal different root inclinations relative to the horizontal and vertical planes. However, after dismantling the specimens after the constant head permeability test, fine surficial roots were also confirmed, as shown in Fig. 4.

After the X-ray CT imaging, all the undisturbed tube samples were carefully saturated through a vacuum saturation method at a prolonged rate, lasting one week. After that, the constant head permeability test (ASTM D2434–19) was carried out to estimate the specimen's saturated hydraulic conductivity (k_{sat}). Three constant heads (i.e., 80 cm, 90 cm, and 100 cm) were used to calculate k_{sat} . Six samples from each burn severity area were tested to obtain the average and representative k_{sat} value for each burn severity. After the test, the roots were carefully

separated from the soil particles for the particle size distribution, specific gravity of soil grains, and root biomass measurement.

2.8. Shear strength test

As the reduced effective stress due to rainfall infiltration is one of the leading causes of shallow landslides, this study measured the shear strength of saturated soils using the undisturbed tube samples collected at a depth of 0–5 cm from the study sites. The direct shear test (ASTM D3080–04) was conducted with the undisturbed samples containing root biomass and solid grains. Upon one-week-long vacuum saturation, specimens were first retrieved from the tube samples using ring knives ($d = 6$ cm, $h = 3$ cm). The interior of the ring knives was coated with petroleum jelly to prevent preferential flow along the perimeter. The

retrieved specimens in ring knives were again submerged in water to ensure saturation. Nine specimens (i.e., three replicates for each normal stress) from each burn severity site were sheared to obtain the drained shear strength parameters with a direct shear testing device (Humboldt HM-5750 Direct/Residual Shear Apparatus). The direct shear tests used three normal stresses (i.e., 50, 100, and 200 kPa) and a constant shear rate of 0.5 mm/min. After the test, the roots were carefully separated from the soil particles to estimate the root biomass.

2.9. Root biomass measurement

This study determined the root biomass contents in the specimens; the root biomass B_R (unit: kg/m³) is defined as the mass of dry root biomass larger than 1 mm divided by its soil volume (Jotisankasa and Sirirattanachai, 2017; Alam et al., 2022). After completion of the direct shear test and constant head permeability test, the specimens were dismantled. Root fragments were separated from the soil particles using a tweezer and soaked in water in a beaker. Afterward, the soaked roots with some impurities were stirred gently for about two minutes, allowing further separation of the soil particle residues and roots. After stirring, the suspended matter was immediately poured into a sieve with an opening size of 1 mm (sieve No. 18). Soil residues passed through sieve No. 18, and roots retained on the sieve were collected. Fine root fragments passed through sieve No. 18 were considered minor and not included in root biomass measurement. This step was repeated several times until all roots were retrieved from the soaked specimen. These filtered and collected roots, which were mainly lengthier and more extensive than 1 mm, were then oven-dried at 50 °C for 48 h and their dry biomass was used for calculating B_R (Blouin et al., 2007).

On the one hand, the soil particles were oven-dried and used to estimate the specific gravity G_s and the gradation of each specimen. The porosity (n) was estimated using Eq. 3 (Jotisankasa and Sirirattanachai, 2017):

$$n = \frac{V_V}{V} = \frac{V - V_S - V_R}{V} = \frac{V - \frac{W_S}{G_s \gamma_w} - \frac{W_R}{G_R \gamma_w}}{V}, \quad (3)$$

where, V_V is the volume of voids, V is the total volume of the specimen, V_S is the volume of soil particles, V_R is the volume of the roots, W_S is the weight of soil particles, W_R is the weight of the roots, γ_w is the unit weight of water, and G_R is the specific gravity of the roots. In this study, G_R was assumed as 0.63 g/cm³ (Foresta et al., 2020).

2.10. Statistical analyses

A paired sample t -test was conducted to assess the normality of the data collected from each laboratory test. This statistical test compared the properties of the ML- and MH-burned soils to those of the UB soils (Table 3). In addition, the coefficient of determination (R^2) was used to evaluate the goodness of fit to estimate the shear strength parameters, cohesion intercept and internal friction angle of soils. The coefficient of determination, R^2 , also assessed the relationship between (1) cohesion intercept and root biomass, (2) saturated hydraulic conductivity and porosity, and (3) porosity and root biomass.

3. Results and analysis

3.1. Soil index properties and root biomass

3.1.1. Dry density ρ_D and specific gravity G_s

The presence of roots in a soil matrix impacts the apparent dry density ρ_D of the soil. Considering that all specimens collected from the near-surface ground have the same volume and assuming the same soil structure, the state of the roots occupying a portion of the soil tends to change the soil's apparent ρ_D . Additionally, the combustion of soil organic matter during forest wildfire contributes to the spatial

variability of the soil's apparent ρ_D . Fig. 5a and Table 4 show the measured ρ_D for test sites. First, the ρ_D of the UB specimens was consistent as 1.27–1.29 g/cm³ throughout the periods. The ρ_D of the ML and MH specimens collected one month after the wildfire was slightly higher than that of UB specimens. Specifically, the mean ρ_D of the ML and MH burned soils was 1.35 g/cm³ and 1.40 g/cm³, respectively, compared to the mean ρ_D of the UB soils of 1.29 g/cm³. However, this tendency disappeared when 6 months elapsed since the wildfire. Within the same burn severity class, the mean ρ_D of the burned soils slightly decreased over time after the wildfire. The mean ρ_D of soils six months after the wildfire decreased from 1.35 g/cm³ to 1.25 g/cm³ in the ML burn severity area and from 1.40 g/cm³ to 1.25 g/cm³ in the MH burn severity area. Herein, the mean ρ_D was computed using 15 samples per test plot and visit, as shown in Table 3. Meanwhile, the soil's apparent ρ_D shows insignificant spatial variability within each burn severity class. Likewise, the specific gravity G_s ranged between 2.64 and 2.70, and the wildfire seemed to hardly influence G_s . However, the statistical results reveal that ρ_D and G_s of the ML- and MH-burned soils significantly differ from those of the unburned soils (i.e., the significance level or p -value < 0.05). The same observation holds for the other tested soil properties: the moisture content, soil gradation, root biomass, shear strength parameters, and saturated hydraulic conductivity substantially differ depending on burn severity and post-wildfire time.

3.1.2. Moisture content w

Fig. 5b and Table 4 also show the natural moisture content w of disturbed soil samples excavated from 0 to 5 cm depth. The w significantly decreased for the two burned areas relative to the UB area one month after the wildfire. The mean w rapidly declined from 17.67 % in the UB site to 11.84 % in the ML site and to 7.44 % in the MH site. This observation is attributable to the moisture evaporation during the

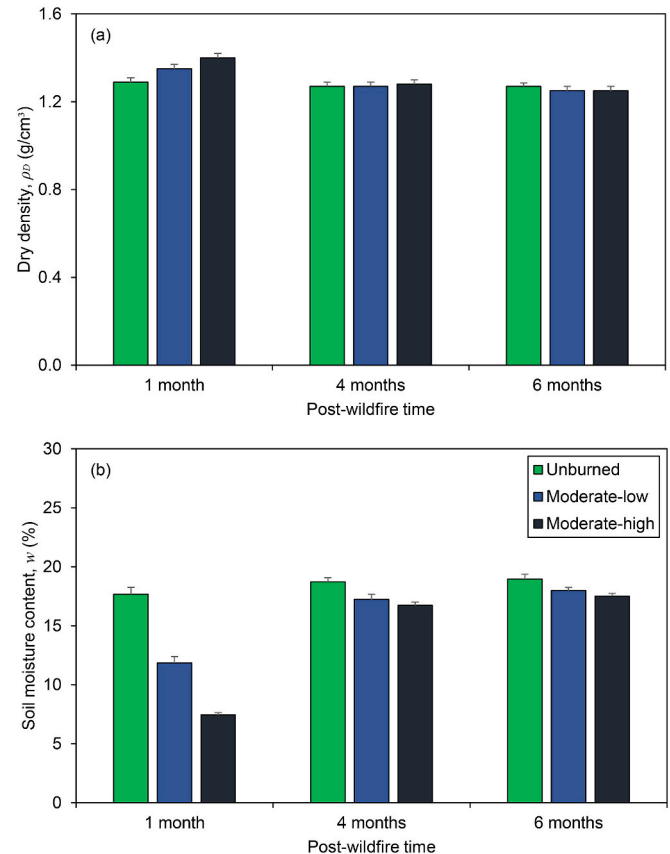


Fig. 5. The temporal variation of (a) apparent soil dry density (ρ_D) and (b) soil moisture content (w) with soil burn severity.

Table 4
Summary of index, geomechanical, and hydraulic properties of burned soils.

Soil burn severity	Unburned (UB)			Moderate-low (ML)			Moderate-high (MH)		
Post-wildfire time (month)	1	4	6	1	4	6	1	4	6
Soil index properties									
Specific gravity, G_s	2.67	2.64	2.66	2.64	2.66	2.70	2.66	2.66	2.67
Dry density, ρ_D (g/cm ³)	1.29	1.27	1.27	1.35	1.27	1.25	1.40	1.28	1.25
Moisture content, w (%)	17.67	18.72	18.95	11.84	17.24	17.98	7.44	16.73	17.51
Gravel content (%)	11.72	12.41	10.66	10.29	12.18	9.33	8.81	10.93	10.42
Sand content (%)	60.82	61.92	62.74	65.72	68.64	76.41	70.31	72.90	77.22
Fines content (%)	27.46	25.68	26.60	23.99	19.18	14.26	20.88	16.17	12.36
Mean grain size, d_{50} (mm)	0.81	0.59	0.79	0.80	0.80	0.80	0.90	0.94	1.28
Liquid limit, LL (%)	32.70	29.41	33.15	28.60	25.32	24.40	28.53	23.59	24.37
Plasticity index, PI (%)	6.45	4.31	4.25	5.68	0.54	0.44	4.79	2.17	3.62
Soil classification (USCS)	SM	SM	SM	SM	SM	SM	SM	SM	SM
Direct shear test results for burned soil samples collected between 0 and 5 cm depth from the ground surface									
Friction angle, ϕ (°)	31.83	30.53	30.47	31.83	33.34	31.07	32.02	33.46	31.42
Cohesion intercept, c_i (kPa)	23.54	22.74	25.68	19.91	17.98	13.36	16.73	15.37	10.30
Root biomass, B_R (kg/m ³)	2.45	2.41	2.67	1.14	0.83	0.48	0.81	0.62	0.37
Constant head permeability test results for burned soil samples collected between 0 and 15 cm depth from the ground surface									
Permeability, k_{sat} ($\times 10^{-5}$ m/s)	4.30	4.51	4.91	2.53	3.74	6.47	1.45	2.87	8.76
B_R (kg/m ³)	3.71	3.64	3.82	2.48	2.27	1.92	2.14	1.85	1.34

wildfire. Moreover, creating a hydrophobic soil layer can slow recovery to a pre-fire moisture level. This condition stayed valid until six months after the wildfire. The w values of the ML and MH sites four and six months after the wildfire were still slightly lower than those of the UB site despite a series of minor rainfall events, although the w difference between the unburned and burned sites is only less than 2 % (Fig. S1 in Supplementary Material). The soil moisture also shows insignificant spatial variability, similar to the apparent ρ_D .

3.1.3. Particle size distribution

Fig. 6 shows the particle size distributions for the soils collected from the three field sampling efforts. Generally, the gravel contents of the soils from the unburned and burned areas (UB, ML, and MH sites) consistently ranged between 9 and 12 %. The wildfire and the elapsed time since the wildfire show no substantial effects on the gravel contents (Table 4). On the other hand, the fines content (i.e., silts and clays)

appears to change with post-wildfire time (Table 4). Specifically, the fines content of the soils collected from the MH site decreased from 20.88 % one month after the wildfire to 12.36 % six months after the wildfire, respectively.

Meanwhile, the fines content at the UB site stayed consistently at 27.46 % one month after the wildfire to 26.6 % six months after. Such changes in the fines content in the ML and MH sites affected their sand contents, which increased with time elapsed since the wildfire. This observation is presumed because of rainfall-induced fines migration or suffusion, where rainwater infiltration washed and displaced fine particles. In the UB site, the fines contents showed insignificant change because live roots protect soils from erosion and migration. Despite the reductions in fines contents over time, the soil classification remained silty sand (SM) for all soils, according to USCS. The soil samples showed a sandy loam to loamy sand texture; it is classified as Arenosols, as defined by WRB.

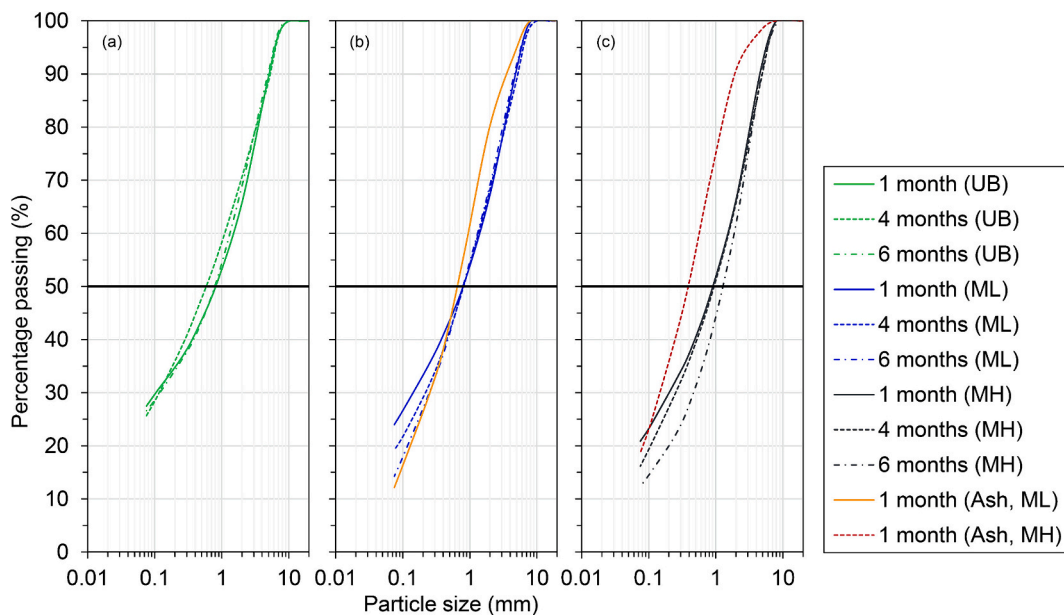


Fig. 6. The particle size distribution of (a) unburned (UB), (b) moderate-low-burned (ML), and (c) moderate-high-burned (MH) soils collected one month, four months, and six months post-wildfire. The particle size distribution of ash samples collected from the ML and MH test plots one-month post-wildfire.

Fig. 6 shows the particle size distributions of the ash deposits collected one month after the wildfire from the ML and MH sites. The MH burn severity produced more ashes than the ML burn severity. Specifically, the ash content was 12.16 % for the ML site and 18.71 % for the MH site. This observation is consistent with the results reported in the literature (Balfour and Woods, 2013). This is plausibly attributed to the transformation of near-surface organic matter to crystalline fragments of calcite and other minerals at higher temperatures, thus, higher burn severity (Bodí et al., 2014).

3.1.4. Root biomass B_R

Fig. 7 shows the variations of B_R with burn severity and post-wildfire time for soil samples retrieved from 0 to 5 cm depth (i.e., those subjected to shear strength test) and from 0 to 15 cm depth (i.e., those subjected to permeability test). As the burn severity increases, the root biomass content of near-surface soils shows a decrease in B_R : $B_R = 3.71\text{--}3.82\text{ kg/m}^3$ in the UB site, $B_R = 1.92\text{--}2.48\text{ kg/m}^3$ in the ML site, and $B_R = 1.34\text{--}2.14\text{ kg/m}^3$ in the MH site for soils from 0 to 15 cm depth. Expectedly, B_R shows a consistent decrease in post-wildfire time because the above-ground tree bodies were burnt, and thus the buried roots decayed with time. Additionally, owing to the high heat caused by the wildfire, the shallower soils contained less root biomass.

3.2. Soil water repellency

Wildfires combust soil organic matter, including vegetation roots and fallen leaves and result in the accumulation of wildfire ashes in near-surface soil layers. This condition causes an increase in soil water repellency (Martins et al., 2020; Akin and Akinleye, 2021; Movasat and Tomac, 2021). The soil water repellency represents the hydrophobicity,

the ability of soil surfaces to resist wetting by water. This study conducted in situ WDPT to assess the influence of the wildfire on the soil water repellency, as shown in Fig. 8. The WDPT results one month after the wildfire clearly show that the water repellency is enhanced with increased burn severity.

Furthermore, the mean WDPTs of ML and MH sites decreased with time: in Site MH, from 203.3 s (strong water repellency) one month after the wildfire to 114.7 s (moderate water repellency) six months after the wildfire, respectively. For Site ML, the WDPT dropped from 82.9 s (moderate water repellency) one month after the wildfire to 39–43 s (weak water repellency) six months after. In contrast, the UB site had non-water-repellent soils with WDPTs of less than 15 s, unaffected by the time elapsed since the wildfire.

In addition to the in situ WDPT tests, the contact angle β was measured with the water droplets made on soil surfaces of laboratory specimens. The contact angle measurement results agreed well with the WDPT results, which confirmed the degree of soil water repellency from in situ measurements, as shown in Fig. 9. The hydrophilicity and hydrophobicity can be categorized into following levels: strongly hydrophilic when $\beta < 30^\circ$, intermediately hydrophilic when $30^\circ < \beta < 60^\circ$, weakly hydrophilic when $60^\circ < \beta < 80^\circ$, neutral when $80^\circ < \beta < 100^\circ$, weakly hydrophobic when $100^\circ < \beta < 120^\circ$, intermediately hydrophobic when $120^\circ < \beta < 150^\circ$, and strongly hydrophobic when $\beta > 150^\circ$. The contact angle of specimens from the MH site was the most significant one month after the forest wildfire (i.e., $\beta = 109^\circ$), indicating that the soil has weak hydrophobicity. In contrast, the soils from the UB site were intermediately hydrophilic with $\beta < 58^\circ$, significantly less than those of the burned areas. The ML-burned soil surface appears weakly hydrophilic to neutral with $74^\circ < \beta < 87^\circ$. Additionally, the degree of hydrophobicity for the burned areas decreased over time, with a higher rate of change for soils with MH burn severity.

3.3. Drained shear strength parameters

Fig. 10 and Table 4 summarized the shear strength parameters obtained for the three sites and for three post-wildfire times. More detailed results can be found in Figs. S2 and S3 in Supplementary Material. Generally, the soils from the wildfire sites (ML and MH sites) showed lower shear strength compared to those from the unburned site (UB site), specifically attributable to the decrease in cohesion intercept (Fig. 10a). As the burn severity increased, the reduction in cohesion intercept became more significant. The cohesion intercept was 23.5 kPa at Site UB, 19.9 kPa at Site ML, and 15.4 kPa at Site MH one month after the wildfire. Such a decrease in the shear strength is primarily because of the reduced soil-root interlocking, as high heat from wildfires caused shrinkage of root biomasses or burned them to ashes. Within the same

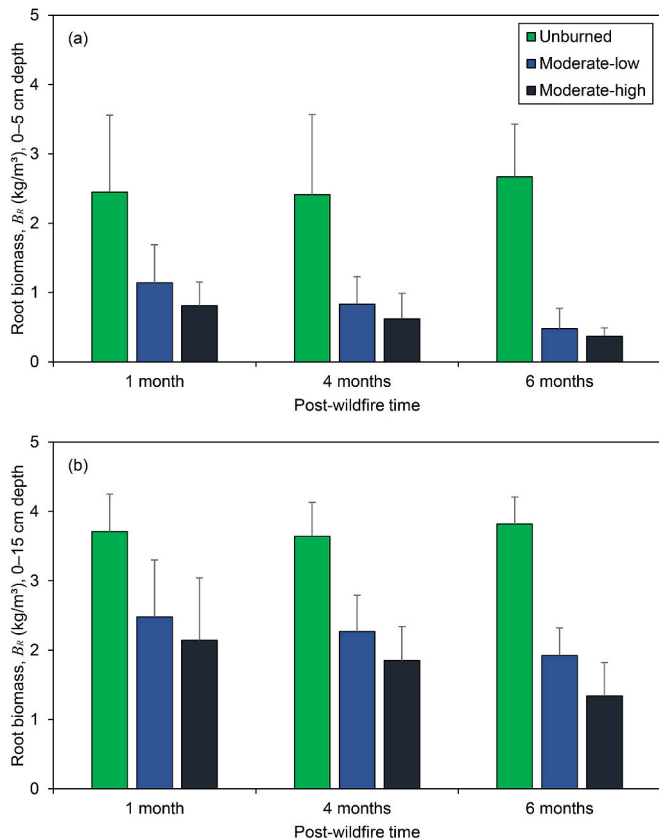


Fig. 7. The temporal variation of root biomass (B_R) with soil burn severity for samples collected (a) from a depth of 0–5 cm (i.e., those samples tested for the direct shear strength) and (b) from a depth of 0–15 cm (i.e., those samples tested for the saturated hydraulic conductivity).

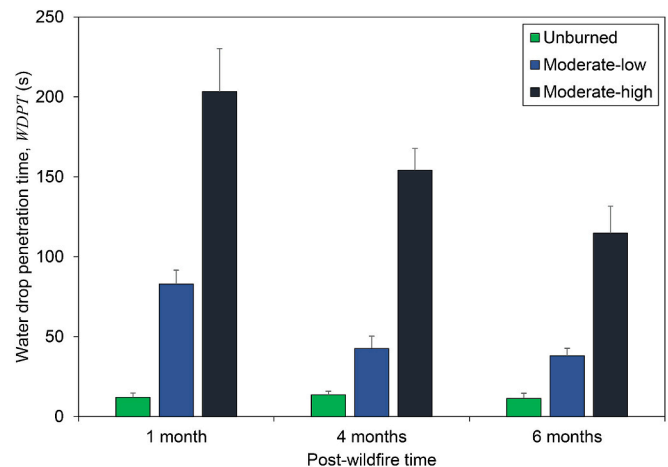


Fig. 8. The temporal variation of soil water repellency with soil burn severity.

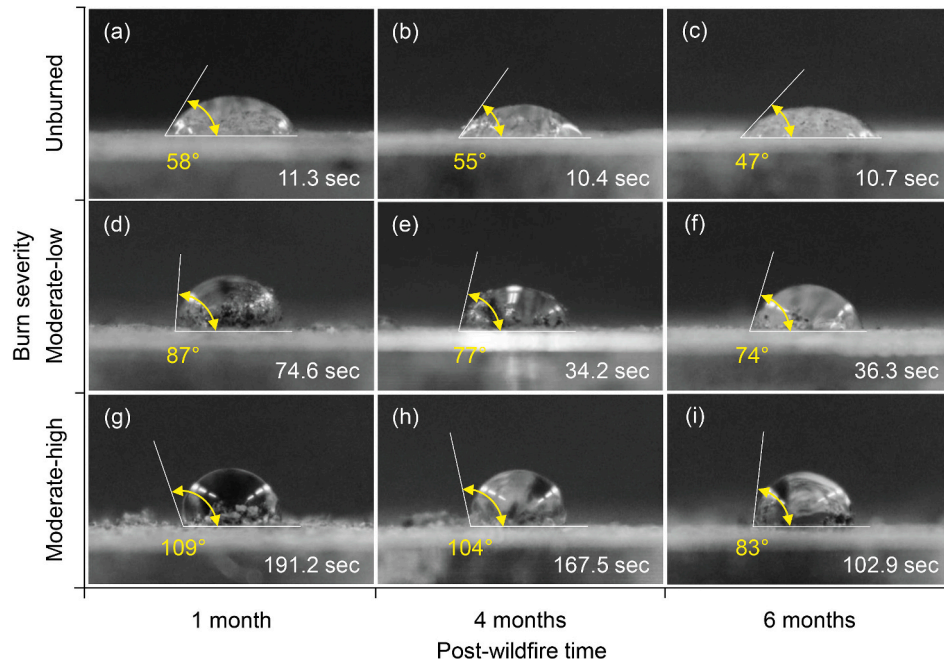


Fig. 9. The temporal variation of contact angle (β) in (a–c) unburned, (d–f) moderate-low-burned, and (g–i) moderate-high-burned soils.

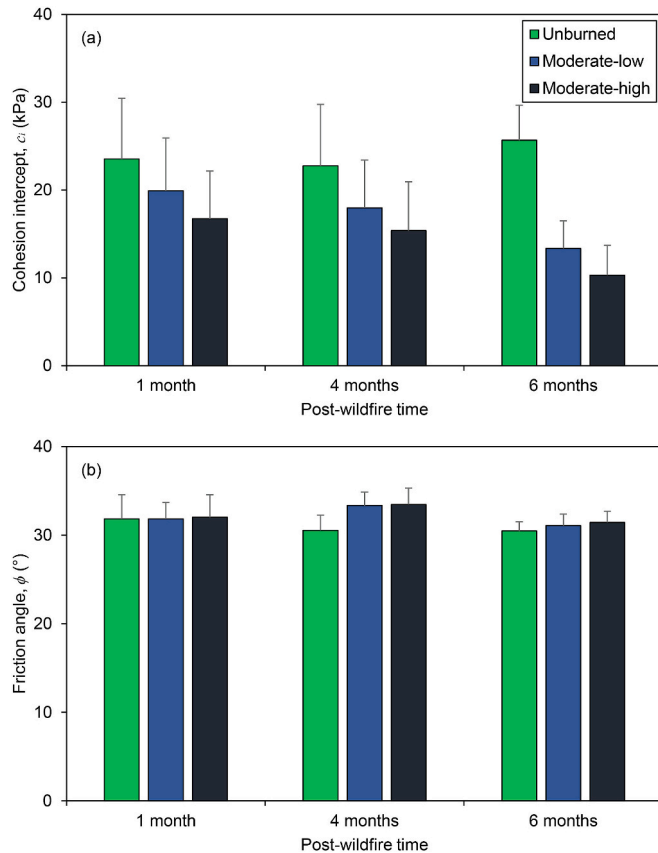


Fig. 10. The temporal variation of (a) cohesion intercept (c_i) and (b) friction angle (ϕ) with soil burn severity.

burn severity class (i.e., ML and MH), the cohesion intercept continuously decreased with time as the root biomasses decayed, and thus, the soil became more loosely packed. This observation is also attributable to the reduced fines content and increased porosity. The cohesion intercept

of the UB soil only showed minimal changes with time, less than 3 kPa, and consistently remained higher than those of the burned soils in all post-wildfire times.

On the other hand, the internal friction angle of soils was not substantially influenced by the soil burn severity and post-wildfire time (Fig. 10b). The absolute difference was less than 3° regardless of the burn severity and post-wildfire time. One note is that we observed some patches of revegetation in the ML and MH sites during the field investigation six months after the wildfire (Fig. 2). However, our result indicates that such revegetation still had limited effect on the recovery of shear strength until six months had elapsed since the wildfire. Instead, root decays appear to have a more profound impact on the continuous decline in cohesion intercept and shear strength over time.

3.4. Saturated hydraulic conductivity

Burn severity and the elapsed post-wildfire time have pronounced effects on k_{sat} of burned soils. Fig. 11 shows the k_{sat} values obtained with

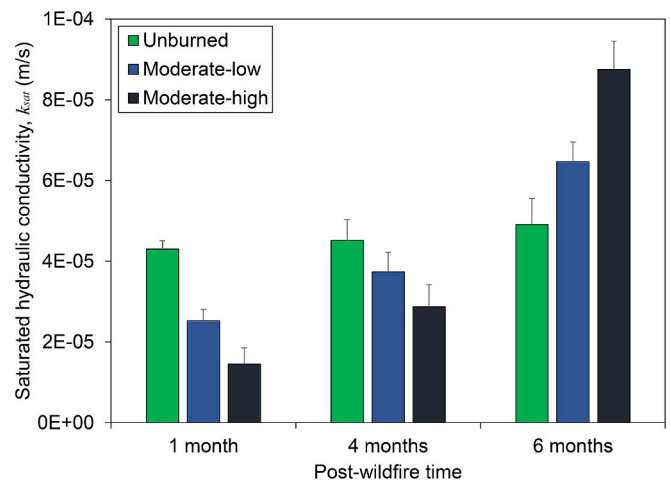


Fig. 11. The temporal variation of saturated hydraulic conductivity (k_{sat}) with soil burn severity.

the tube samples from the test sites at different post-wildfire times. One month after the wildfire, the mean k_{sat} of the ML and MH sites was 2.5×10^{-5} m/s and 1.5×10^{-5} m/s, respectively, significantly lower than that of the UB site (i.e., $k_{sat} = 4.3 \times 10^{-5}$ m/s). This observation is attributable to the ashes deposited at the top portion of the ML and MH soils.

Meanwhile, the mean k_{sat} of the burned soils increased with time and became higher than that of the UB site six months after the wildfire. Specifically, at this post-wildfire time, the mean k_{sat} of the ML and MH sites was 6.5×10^{-5} m/s and 8.8×10^{-5} m/s, respectively. This observation is again attributable to the decays of roots, which created extensive flow paths for hydraulic conductivity and also owing to the removal of fines. The mean k_{sat} of the UB site stayed consistent until six months after the wildfire, although it slightly increased from 4.3×10^{-5} m/s to 4.9×10^{-5} m/s.

4. Discussion and implications

4.1. Particle size distribution temporal change

This study reports no substantial differences (i.e., $< 5\%$) in the gravel fraction within- and across-site for each burn severity class and from the three post-wildfire times. In contrast, there is a relatively appreciable increase in the sand fraction and decrease in fines content in the ML and MH sites, precisely four and six months after the wildfire relative to the UB site (i.e., $> 5\%$ increase or decrease). This observation is consistent with the particle size distribution shift over time reported by Lei et al. (2022). Some studies have shown that burning and (or) heating of soils under high temperatures tend to aggregate fine particles, specifically clay particles, into coarser particles, thus increasing the sand fraction (e.g., Arunrat et al., 2021; Fachin et al., 2021). Accordingly, the appreciation of sand fraction and the depreciation of fines content of burned soils from the two burn severity classes observed one month after the wildfire can also be solely explained by mineralogical assemblage and the thermal fusion of clay particles. Meanwhile, the particle size distribution shift over time was attributable to the selective removal of fines content on the uppermost surface of each burned site through fines migration and erosion caused by wind and (or) rainwater. Aeolian sediment transport through wind erosion post-wildfire has been reported in the literature to cause on-site soil loss (Jeanneau et al., 2019). Likewise, redistribution of near-surface sediments usually occurs, particularly after the loss of vegetation covers and plant litter layers due to wildfires following variable natural rainfall events (Hueso-González et al., 2018; Biswas et al., 2021).

4.2. Soil water repellency temporal change

Our WDPT test results above show that the burned test plots exhibit different degrees of soil water repellency depending on burn severity and post-wildfire time. Plausibly, given the hydrophilic characteristics of soils from the UB site, the burned soils developed hydrophobicity post-wildfire. Stronger hydrophobicity was observed for the MH-burned soils than for the ML-burned and UB soils, consistent with observations from previous studies (Glenn and Finley, 2010; Lei et al., 2022). Creating hydrophobic soil layers following wildfires can significantly result in more pronounced surface-runoff discharges due to rainfall infiltration inhibition. As discussed in Section 4.1, fine sediments on the uppermost soil layer can be mobilized and redistributed spatially and temporally due to these runoffs or overland flows (i.e., flooding) during rainstorms with varying precipitation intensity. Gabet (2003) demonstrated through a simple field experiment how a prescribed fire led to the creation and deposition of hydrophobic layers beneath the surface (i.e., 1–2 cm) of previously hydrophilic soil before burning. In this study, the soil water repellency of both burned sites persisted for six months after the wildfire but weakened over time. This spatiotemporal weakening of soil water repellency agrees with the observations by Huffman et al. (2001), who reported that fire-induced soil hydrophobicity persisted for

at least 22 months.

4.3. Shear strength temporal change

The results on drained shear strength reveal that the amount of root biomass, B_R , substantially influences the cohesion intercept. The cohesion intercept positively correlates to B_R with a power relationship, as shown in Fig. 12. More importantly, Fig. 12 highlights the impact of burn severity and post-wildfire time on the B_R degradation, which causes cohesion intercept reduction. This result is consistent with previous studies that reported a positive correlation between the root fraction and the cohesion intercept from the Mohr-Coulomb shear strength criterion for unburned soils (Saifuddin et al., 2015; Jotisankasa et al., 2018).

Recently, previous studies have investigated the effect of burn severity and post-wildfire time on the shear strength of burned soils and compared it to the shear strength of UB soils (Lei et al., 2022; Peduto et al., 2022a; Peduto et al., 2022b). Generally, these studies reported a reduction in the shear strength of field-sampled burned soils. The reduction in the shear strength was attributed to the decrease in the cohesion intercept provided by the root system spatially (i.e., related to burn severity) and temporally (i.e., related to post-wildfire time). This report is consistent with the results obtained in the presented study. Specifically, Lei et al. (2022) reported that the cohesion intercept in the high-burned area was reduced relative to the UB area by approximately 10 kPa, 12.5 kPa, and 17 kPa two months, one year, and two years after the forest wildfire, respectively. Their study also reported a less pronounced difference temporally in the shear strength of low-burned and unburned areas (i.e., < 1.3 kPa). Peduto et al. (2022a) and Peduto et al. (2022b) reported a reduction in the cohesion intercept by ~ 1.4 kPa with a low/moderate-to-low burn severity five months after a 2019 wildfire in Southern Italy. Likewise, one week after a 2020 wildfire in Southern Italy, the cohesion intercept of a moderate-high/high burned area decreased by 1.4 kPa relative to the UB area (Peduto et al., 2022b). The reductions in the cohesion intercept observed in this study for the MH site relative to the UB site were comparable to those reported by Lei et al. (2022), even at shorter post-wildfire times.

Besides the burn severity, the decomposition rate of buried roots and revegetation rate after exposure to wildfire heavily affect the shear strength of root-soil matrices, particularly the cohesion intercept (Preti, 2013). Peduto et al. (2022b) observed that the cohesion intercept of undisturbed samples from a moderate-high/high burned area eight months after a 2020 wildfire in Southern Italy slightly increased by 2

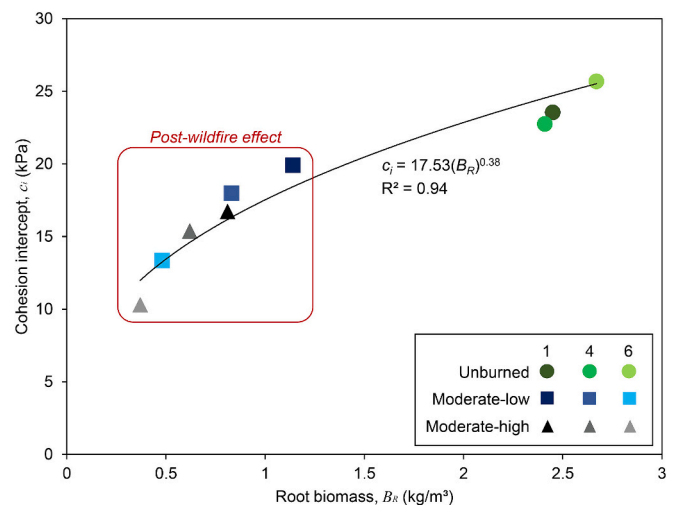


Fig. 12. The relationship between c_i and B_R highlighting the effect of soil burn severity (i.e., unburned, moderate-low, and moderate-high) and post-wildfire time (i.e., one month, four months, and six months).

kPa compared to one week after the wildfire due to the revegetation they observed from that specific test plot. This study also observed such revegetation with irregular patches over the burned and UB areas during the second and third field visits. However, our burned test plots were less influenced by the revegetation; thus, there was no cohesion intercept recovery four and six months after the wildfire. In contrast, the seasonal vegetation cycle in the UB test plot was more significant than those in the burned areas, which explains the increase in cohesion intercept by 2 kPa in the UB samples collected six months after the wildfire during the third field visit in September 2022 (Fig. 10).

Meanwhile, the experiment results from this study and previous studies suggest that the wildfire and post-wildfire time have only minimal impact on the internal friction angle with less than a 3° difference. In contrast, the internal friction of soils burned in a muffle furnace at 600 °C was lower by approximately 4 °C than those burned at 300 °C and naturally burned in the field during an actual forest wildfire (Peduto et al., 2022a). Interestingly, the cohesion intercept of these muffle furnace-burned soils remained unchanged. This observation implies that substantially higher temperatures are required to cause significant changes in grain size, fabric, and texture of soils, hence changing the internal friction angle.

4.4. Saturated hydraulic conductivity temporal change

Wildfires tremendously alter watersheds' landscapes depending on burn severity, duration, and frequency. Thus, the hydraulic response of the watershed after exposure to wildfires and immediately subjected to variable precipitations can vary substantially spatially and temporally. Several studies showed that burned soils' saturated hydraulic conductivity k_{sat} considerably decreased relative to unburned soils (Nyman et al., 2011; Holden et al., 2014; Varela et al., 2015; Ebel and Moody, 2020). However, only a few studies have correlated the k_{sat} reduction with soil burn severity (McGuire et al., 2021) and post-wildfire time (Lei et al., 2022). The results on the k_{sat} reduction immediately after the wildfire agree with these previous studies, indicating the post-wildfire short-term effect, as depicted in Fig. 13. Such k_{sat} reduction immediately after the wildfire is mainly due to the fine ash clogging in soil pores

and partly attributable to wildfire-induced or enhanced soil water repellency (Wieting et al., 2017; Lei et al., 2022).

This study also observed long-term gains in k_{sat} for all burned soils four and six months after the wildfire (Fig. 13). These k_{sat} increases over months can be explained by the long-term post-wildfire effect, mainly root decay and ensuing fines migration, which enhances pore connectivity and porosity n . Fig. 14a depicts the k_{sat} - n relation, and Fig. 14b shows the n - B_R relationship. This result highlights the combined effect of soil burn severity and post-wildfire time on the surficial root decay process, leading to its complete impoverishment. As the root biomass B_R decreases with time due to decay, n and thus k_{sat} increase accordingly. This observation in the study agrees well with that reported by Jotisankasa and Sirirattanachai (2017), who, however, did not consider the wildfire effect. Decay of burned roots increases porosity and creates pathways for water flow within the soil sample, as shown in Fig. 15. Vegetation and live roots provide an excellent natural solution for soil erosion mitigation against groundwater flows. However, as the root decays, precipitation and rainfall infiltration can cause the migration of fine ash and soil particles. Therefore, as observed in this study, the decrease in fines content of all burned soils can also contribute to the substantial gain in k_{sat} over time (Fox et al., 2007; Lei et al., 2022).

4.5. Practical implications to post-wildfire landslides

Global research efforts have focused on understanding factors contributing to shallow landslides and debris flows in forested, mountainous regions. However, the geomechanical and hydraulic properties of slopes affected by wildfires, specifically the influence of soil burn severity and post-wildfire temporal changes, have yet to receive much attention. Limited literature exists on how soil index properties, mechanical attributes, and hydrologic characteristics evolve temporally post-wildfire, particularly concerning spatial variations in soil burn severity. This study provides a unique opportunity to advance our understanding of dynamic post-wildfire hydrological and geomorphological processes.

This study highlights the dynamic changes in grain size distributions, driven by erosion and mobilization of fine sediments on the uppermost

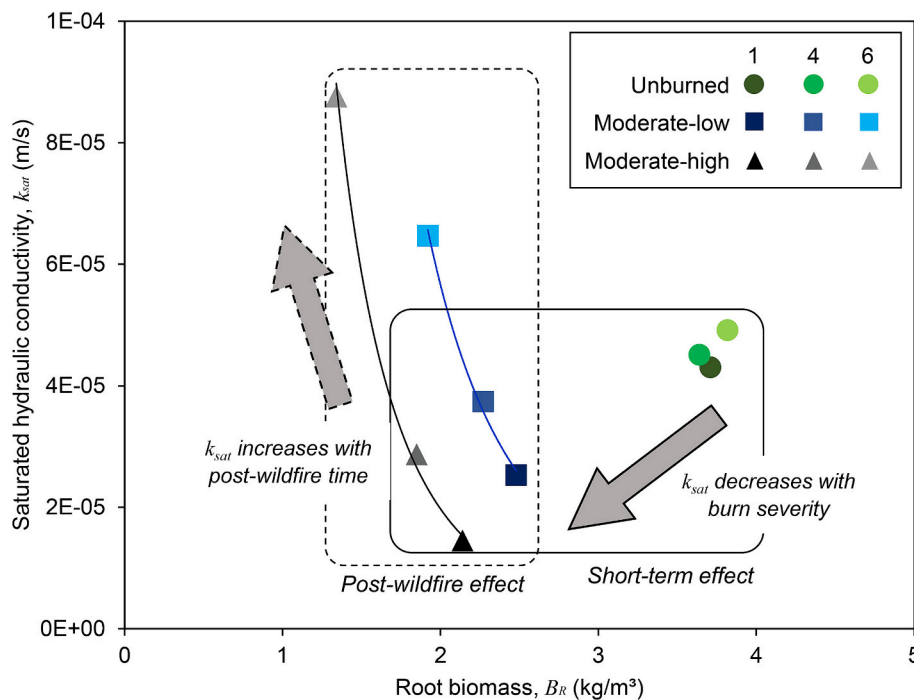


Fig. 13. The relationship between k_{sat} and B_R highlighting the effect of soil burn severity (i.e., unburned, moderate-low, and moderate-high) and post-wildfire time (i.e., one month, four months, and six months).

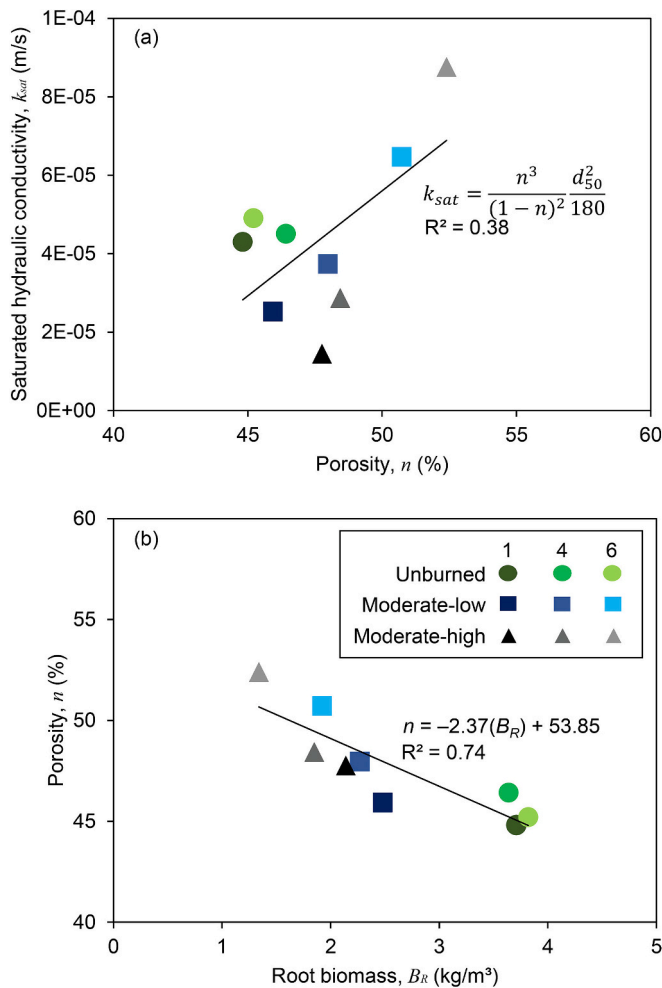


Fig. 14. (a) The relationship between k_{sat} and porosity (n) and (b) the relationship between n and B_R highlighting the effect of soil burn severity (i.e., unburned, moderate-low, and moderate-high) and post-wildfire time (i.e., one month, four months, and six months).

soil layer post-wildfire. This sequential process, involving the formation of a hydrophobic layer beneath the soil surface, hinders rainwater infiltration and exacerbates overland flows, potentially leading to slope failures like debris flows. Additionally, this study demonstrates that fine ash particles from burned vegetation and plant litter can clog pores in the uppermost soil layer, reducing the saturated hydraulic conductivity k_{sat} of the underlying parent soil in the initial four months after the wildfire. The appreciation of k_{sat} observed six months post-wildfire is directly linked to root decomposition and fines removal, creating water pathways that facilitate preferential or macropore flow through the soil sample. Consequently, improving hydrologic model parameter estimation for post-wildfire surface runoff, infiltration, and sediment transport requires consideration of the combined effects of burn severity and post-wildfire time on watersheds (Nyman et al., 2014; Ebel and Moody, 2020; Perkins et al., 2022).

This study reveals that the decomposition rate of surficial fine roots is closely tied to soil burn severity. Decaying and dead roots contribute to a decline in the shear strength of burned soils, primarily driven by reduced cohesion intercept due to weakened soil suction. Simultaneously, lower tensile strength from deteriorating roots diminishes their mechanical reinforcement through soil anchoring (Schwarz et al., 2010). These findings underscore the importance of considering post-wildfire time effects on the evolution of geomechanical and hydrologic characteristics in burned soils for more accurate slope stability analysis (McGuire et al., 2016; Gonzalez-Ollauri and Mickovski, 2017).

This study proves that soil burn severity and post-wildfire time impact burned soil's shear strength and hydraulic behaviors. Thus, the documented results herein may be incorporated into local infiltration and stability analyses of wildfire-affected slopes (i.e., using conceptual, analytical, and numerical models), landslide susceptibility and hazard mapping, and the design of mitigation or protection measures. Meanwhile, the results presented in this study can be site-sensitive and time-dependent; thus, further study is required to extend to the watershed scale and the longer time scale. Remarkably, long-term investigation and spatiotemporal monitoring of broader and more various post-wildfire areas before the burned watershed has recovered to its pre-wildfire state will be invaluable in deriving more generic hydromechanical relations.

Although this study has attempted to conduct sophisticated soil sampling and laboratory testing with care, it lacks a field-testing methodology that can determine hydromechanical properties in the field more directly. Advancements in field test equipment and remote sensing technology will provide richer information, including root characterization and decay modeling, root distribution, and spatiotemporal analysis of vegetation recovery and ash and fine sediment transport post-wildfire.

5. Conclusion

This study elucidates the changes in burned soils' index properties and geomechanical and hydraulic characteristics after the March 2022 wildfire in Uljin County, South Korea. Specifically, the results provide insights into the combined effect of soil burn severity and post-wildfire time on the observed spatiotemporal shift of these critical soil engineering properties. The results presented in this study are site-sensitive and time-dependent; thus, they may only be acceptable to the test-plot scale and not on the watershed scale of the study region. The main findings of this study are as follows:

- The continuous deterioration of fine surficial roots highly depended on the soil burn severity. Throughout three field sampling efforts post-wildfire, the B_R of the ML and MH-burned soils were consistently substantially lower than those of the UB soils. The root's deterioration significantly reduced the soil's c_i temporally, with ϕ significantly unaffected. The burned soil's c_i was 1.80–2.30 times lower than the UB soil six months post-wildfire.
- One month and four months post-wildfire, the burned soil's k_{sat} was considerably reduced by approximately 1.22–3.15 times from the pre-wildfire condition. The observed k_{sat} reduction was due to the fine ash-clogged pores and the persistent presence of hydrophobic layers immediately beneath the soil surface. However, six months post-wildfire, the burned soil's k_{sat} rapidly increased by approximately twice that of the pre-wildfire condition because of macropore flow passages formed by impoverished surficial fine roots.
- The sand fraction appreciation and fines content depreciation and the continuous weakening of soil water repellency over the investigation period have also emphasized their role in the observed temporal shifts on the geomechanical and hydraulic properties of the burned soils. The wildfire undoubtedly modified the landscape and enhanced the externally naturally driven erosive processes due to wind and rainwater due to the removal of vegetation cover and plant litter.

CRediT authorship contribution statement

Ryan A. Ramirez: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Woojae Jang:** Investigation, Resources, Writing – review & editing. **Tae-Hyuk Kwon:**

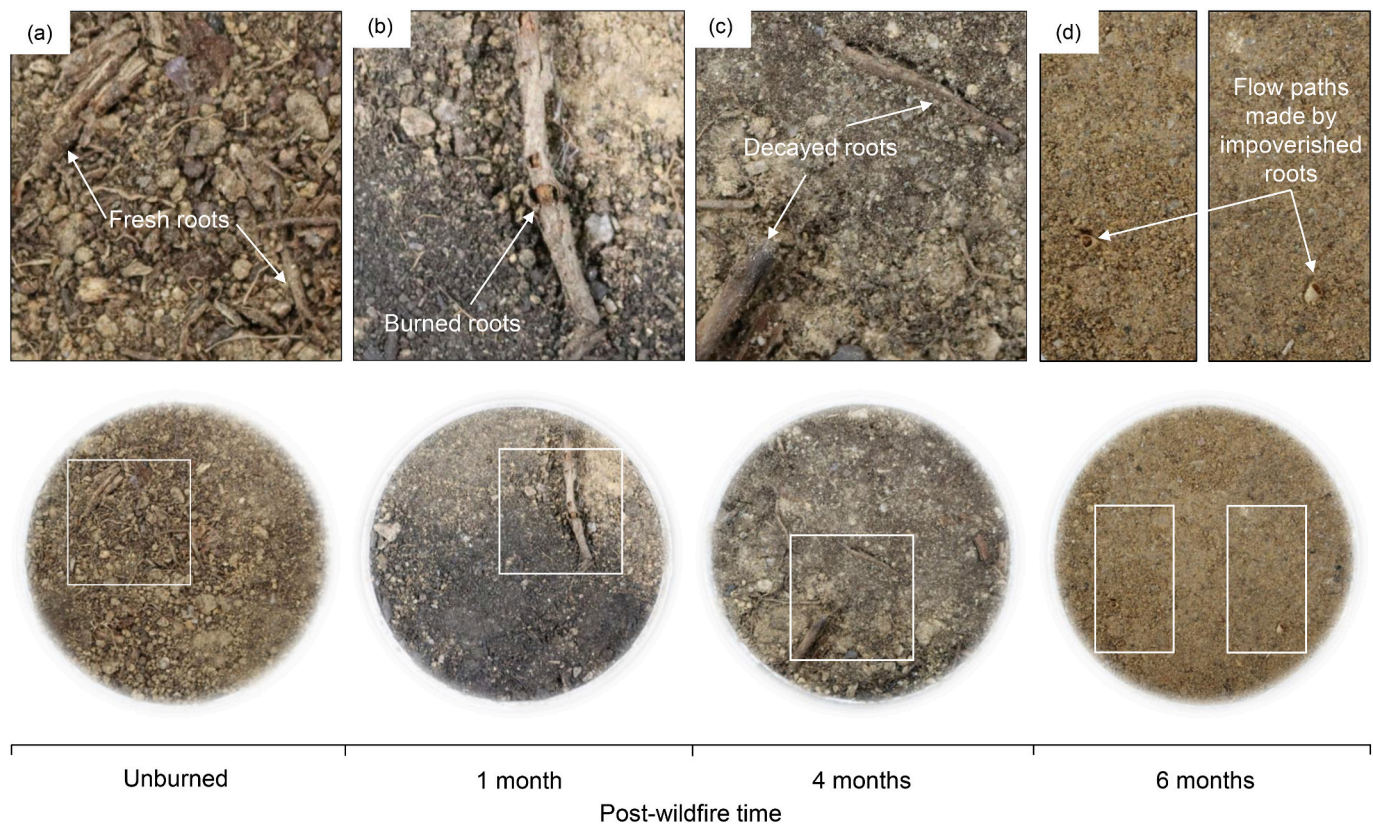


Fig. 15. The photos showing the surficial root biomass state for the (a) unburned soil and moderate-high-burned soils, (b) one month, (c) four months, and (d) six months post-wildfire.

Conceptualization, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgements

This work was financially supported by Korea Ministry of Land, Infrastructure and Transport (MOLIT) as “Innovative Talent Education Program for Smart City” and by Korea Electric Power Corporation (Grant number: R22XO05-05). R. Ramirez was supported by the National Research Council of the Philippines (NRCP) through the Support to Research Dissemination in Local and International Platforms (RDLIP) component of the Basic Research Information Translation for Empowerment in the Regions (BRITER) Program. The authors thank the European Space Agency (ESA) for freely supplying Sentinel-2 Multispectral Instrument (MSI) images used to produce the soil burn severity map of the study area.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geodrs.2024.e00856>.

References

- Abbate, A., Longoni, L., Ivanov, V.I., Papini, M., 2019. Wildfire impacts on slope stability triggering in mountain areas. *Geosciences* 9 (10), 417.
- Akin, I.D., Akinleye, T.O., 2021. Water vapor sorption behavior of wildfire-burnt soil. *J. Geotech. Geoenviron. Eng.* 147 (11), 04021115.
- Alam, M., Jiang, Y.J., Umar, M., Su, L.J., Rahman, M., Ullah, F., 2022. Influence of drainage and root biomass on soil mechanical behavior in triaxial tests. *Acta Geotech.* 17, 2875–2893.
- Arunrat, N., Sreenonchai, S., Hatano, R., 2021. Impact of burning on soil organic carbon of maize-upland rice system in Mae Chaem Basin of Northern Thailand. *Geoderma* 392, 115002.
- Balfour, V.N., Woods, S.W., 2013. The hydrological properties and the effects of hydration on vegetative ash from the Northern Rockies, USA. *Catena* 111, 9–24.
- Biswas, T.K., Karim, F., Kumar, A., Wilkinson, S., Guerschman, J., Rees, G., McInerney, P., Zampatti, B., Sullivan, A., Nyman, P., Sheridan, G.J., Joehnk, K., 2021. 2019–2020 bushfire impacts on sediment and contaminant transport following rainfall in the upper Murray River catchment. *Integr. Environ. Assess. Manag.* 17 (6), 1203–1214.
- Blouin, M., Barot, S., Roumet, C., 2007. A quick method to determine root biomass distribution in diameter classes. *Plant Soil* 290, 371–381.
- Bodí, M.B., Martín, D.A., Balfour, V.N., Santín, C., Doerr, S.H., Pereira, P., Cerdà, A., Mataix-Solera, J., 2014. Wildland fire ash: production, composition and eco-hydro-geomorphic effects. *Earth Sci. Rev.* 130, 103–127.
- Cannon, S.H., Bigio, E.R., Mine, E., 2001. A process for fire-related debris flow initiation, Cerro Grande fire, New Mexico. *Hydrol. Process.* 15 (15), 3011–3023.
- Cui, Y., Cheng, D., Chan, D., 2019. Investigation of post-fire debris flows in Montecito. *ISPRS Int. J. Geo Inf.* 8 (1), 5.
- Ebel, B.A., Moody, J.A., 2020. Parameter estimation for multiple post-wildfire hydrologic models. *Hydrol. Process.* 34 (21), 4049–4066.
- Fachin, P.A., Costa, Y.T., Thomaz, E.L., 2021. Evolution of the soil chemical properties in slash-and-burn agriculture along several years of fallow. *Sci. Total Environ.* 764, 142823.
- Foresta, V., Capobianco, V., Cascini, L., 2020. Influence of grass roots on shear strength of pyroclastic soils. *Can. Geotech. J.* 57 (9), 1320–1334.
- Fox, D.M., Darboux, F., Carrega, P., 2007. Effects of fire-induced water repellency on soil aggregate stability, splash erosion, and saturated hydraulic conductivity for different size fractions. *Hydrol. Process.* 21 (17), 2377–2384.
- Gabet, E.J., 2003. Post-fire thin debris flows: sediment transport and numerical modelling. *Earth Surf. Process. Landf.* 28 (12), 1341–1348.

- Glenn, N.F., Finley, C.D., 2010. Fire and vegetation type effects on soil hydrophobicity and infiltration in the sagebrush-steppe: I. Field analysis. *J. Arid Environ.* 74 (6), 653–659.
- Gonzalez-Ollauri, A., Mickovski, S.B., 2017. Hydrological effect of vegetation against rainfall-induced landslides. *J. Hydrol.* 549, 374–387.
- Holden, J., Wearing, C., Palmer, S., Jackson, B., Johnston, K., Brown, L.E., 2014. Fire decreases near-surface hydraulic conductivity and macropore flow in blanket peat. *Hydrol. Process.* 28 (5), 2868–2876.
- Hueso-González, P., Martínez-Murillo, J.F., Ruiz-Sinoga, J.D., 2018. Prescribed fire impacts on soil properties, overland flow and sediment transport in a Mediterranean forest: a 5 year study. *Sci. Total Environ.* 636, 1480–1489.
- Huffman, E.L., MacDonald, L.H., Stednick, J.D., 2001. Strength and persistence of fire-induced soil hydrophobicity under ponderosa and lodgepole pine, Colorado Front Range. *Hydrol. Process.* 15 (15), 2877–2892.
- Inbar, A., Lado, M., Sternberg, M., Tenau, H., Ben-Hur, M., 2014. Forest fire effects on soil chemical and physicochemical properties, infiltration, runoff, and erosion in a semiarid Mediterranean region. *Geoderma* 221–222, 131–138.
- IUSS Working Group WRB, 2022. World reference base for soil resources. In: International Soil Classification System for Naming Soils and Creating Legends for Soil Maps, 4th Ed. 2022. International Union of Soil Sciences, Vienna, Austria.
- Jeanneau, A.C., Ostendorf, B., Herrmann, T., 2019. Relative spatial differences in sediment transport in fire-affected agricultural landscapes: a field study. *Aeolian Res.* 39, 13–22.
- Jotisankasa, A., Sirirattanachai, T., 2017. Effects of grass roots on soil-water retention curve and permeability function. *Can. Geotech. J.* 54 (11), 1612–1622.
- Jotisankasa, A., Taworn, D., Chumchey, N., Sukolrat, J., 2018. Quantification of root reinforcement in bio-slope stabilization: laboratory and field studies. In: MATEC Web of Conferences 195, 03002.
- Jung, S.H., Lim, C.H., Kim, A.R., Woo, D.M., Kwon, H.J., Cho, Y.C., Lee, C.S., 2017. Edge effects confirmed at the clear-cut area of Korean red pine forest in Uljin, eastern Korea. *J. Ecol. Environ.* 41, 36.
- Kean, J.W., Staley, D.M., Lancaster, J.T., Rengers, F.K., Swanson, B.J., Coe, J.A., Hernandez, J.L., Sigman, A.J., Allstadt, K.E., Lindsay, D.N., 2019. Inundation, flow dynamics, and damage in the 9 January 2018 Montecito debris-flow event, California, USA: opportunities and challenges for post-wildfire risk assessment. *Geosphere* 15 (4), 1140–1163.
- Key, C.H., Benson, N.C., 2006. Landscape assessment (LA) sampling and analysis methods. In: USDA Forest Service-General Technical Report RMRS-GTR, 164 RMRS-GTR.
- Korea Forest Service (KFS), 2022. Data was accessed on April 4, 2022 from <https://www.forest.go.kr/kfswb/kfs/idx/Index.do>.
- Korea Meteorological Administration (KMA), 2022. Data was accessed on April 4, 2022 from <https://www.kma.go.kr/neng/index.do>.
- Lei, M., Cui, Y., Ni, J., Zhang, G., Li, Y., Wang, H., Liu, D., Yi, S., Jin, W., Zhou, L., 2022. Temporal evolution of the hydromechanical properties of soil-root systems in a forest fire in China. *Sci. Total Environ.* 809, 151165.
- León, J., Echeverría, M.T., Martí, C., Badía, D., 2015. Can ash control infiltration after burning? An example in burned calcareous and gypseous soils in the Ebro Basin (NE Spain). *Catena* 135, 377–382.
- Martins, M.A.S., Verheijen, F.G.A., Malvar, M.C., Serpa, D., González-Pelayo, O., Keizer, J.J., 2020. Do wildfire and slope aspect affect soil water repellency in eucalypt plantations? – a two-year high resolution temporal dataset. *Catena* 189, 104471.
- McGuire, L.A., Rengers, F.K., Kean, J.W., Coe, J.A., Mirus, B.B., Baum, R.L., Godt, J.W., 2016. Elucidating the role of vegetation in the initiation of rainfall-induced shallow landslides: insights from an extreme rainfall event in the Colorado front range. *Geophys. Res. Lett.* 43 (17), 9084–9092.
- McGuire, L.A., Youberg, A.M., Rengers, F.K., Abramson, N.S., Ganesh, I., Gorri, A.N., Hoch, O., Johnson, J.C., Lamom, P., Prescott, A.B., Zanetell, J., Fenerty, B., 2021. Extreme precipitation across adjacent burned and unburned watersheds reveals impacts of low severity wildfire on debris-flow processes. *J. Geophys. Res. Earth* 126 (4), e2020JF005997.
- Moody, J.A., Martin, D.A., Haire, S.L., Kinner, D.A., 2008. Linking runoff response to burn severity after a wildfire. *Hydrol. Process.* 22 (13), 2063–2074.
- Movasat, M., Tomac, I., 2021. Assessment of physical properties of water-repellent soils. *J. Geotech. Geoenviron. Eng.* 147 (9), 06021010.
- Nuraje, N., Khan, W.S., Lei, Y., Ceylan, M., Asmatulu, R., 2013. Superhydrophobic electrospun nanofibers. *J. Mater. Chem. A* 1 (6), 1929–1946.
- Nyman, P., Sheridan, G.J., Smith, H.G., Lane, P.N.J., 2011. Evidence of debris flow occurrence after wildfire in upland catchments of south-East Australia. *Geomorphology* 125 (3), 383–401.
- Nyman, P., Sheridan, G.J., Smith, H.G., Lane, P.N.J., 2014. Modeling the effects of surface storage, macropore flow and water repellency on infiltration after wildfire. *J. Hydrol.* 513, 301–313.
- Oh, I.H., Yang, S.J., Heo, C.H., Lee, J.H., Kim, E.J., Cho, S.J., 2022. Study on the controlling factors of Li-bearing pegmatite intrusions for mineral exploration, Uljin, South Korea. *Minerals* 12 (5), 589.
- Palmer, J., 2022. The one-two punch of fires and mudslides. *Nature* 601, 184–186.
- Parks, S.A., Dillon, G.K., Miller, C., 2014. A new metric for quantifying burn severity: the relativized burn ratio. *Remote Sens.* 6 (3), 1827–1844.
- Peduto, D., Iervolino, L., Foresta, V., 2022a. Experimental analysis of the fire-induced effects on the physical, mechanical, and hydraulic properties of sloping pyroclastic soils. *Geosciences* 12 (5), 198.
- Peduto, D., Iervolino, L., Esposito, G., Foresta, V., Matano, F., Masi, R., 2022b. Clues of wildfire-induced geotechnical changes in volcanic soils affected by post-fire slope instabilities. *Bull. Eng. Geol. Environ.* 81, 454.
- Perkins, J.P., Diaz, C., Corbett, S.C., Cerovski-Darriau, C., Stock, J.D., Prancevic, J.P., Micheli, E., Jasperse, J., 2022. Multi-stage soil-hydraulic recovery and limited ravel accumulations following the 2017 Nuns and Tubbs wildfires in Northern California. *J. Geophys. Res. Earth* 127 (6), e2022JF006591.
- Preti, F., 2013. Forest protection and protection forest: tree root degradation over hydrological shallow landslides triggering. *Ecol. Eng.* 61 (Part C), 633–645.
- Quintano, C., Fernández-Manso, A., Fernández-Manso, O., 2018. Combination of Landsat and Sentinel-2 MSI data for initial assessing of burn severity. *Int. J. Appl. Earth Obs. Geoinf.* 64, 221–225.
- Saifuddin, M., Osman, N., Rahman, M.M., Boyce, A.N., 2015. Soil reinforcement capability of two legume species from plant morphological traits and mechanical properties. *Curr. Sci.* 108 (7), 1340–1347.
- Schwarz, M., Cohen, D., Or, D., 2010. Root-soil mechanical interactions during pullout and failure of roots bundles. *J. Geophys. Res. Earth* 115 (F4), F04035.
- Varela, M.E., Benito, E., Keizer, J.J., 2015. Influence of wildfire severity on soil physical degradation in two pine forest stands of NW Spain. *Catena* 133, 342–348.
- Vieira, D.C.S., Malvar, M.C., Martins, M.A.S., Serpa, D., Keizer, J.J., 2018. Key factors controlling the post-fire hydrological and erosive response at micro-plot scale in a recently burned Mediterranean forest. *Geomorphology* 319, 161–173.
- Wieting, C., Ebel, B.A., Singha, K., 2017. Quantifying the effects of wildfire on changes in soil properties by surface burning of soils from the Boulder Creek critical zone observatory. *J. Hydrol. Region. Stud.* 13, 43–57.